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Problem Chosen

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Summary Sheet

Evaluating and Recommending SDEs for the 2032 Brisbane Olympics

Summary

The International Olympic Committee (IOC) is planning the 2032 Summer Olympics in Brisbane, Australia, with the goal of keeping the Games relevant and impactful by adding or removing sports, disciplines, or events (SDEs) that resonate with modern values and appeal to a global audience. We developed a comprehensive mathematical model to evaluate SDEs against the IOC's established criteria.

For Task 1, we pre-processed and visualized the large amount of data collected and built model assumptions based on the constraints of the actual problem to solve it. **17 secondary indicators** related to the Olympic SDEs were selected from the six indicators of **Gender equality**, **Safety and fair play**, **Sustainability**, **Inclusiveness**, **Relevance and innovation**, **Visibility and feasibility**, and then use the entropy weight method to calculate the weight of these indicators and make rankings.

For Task 2, the final weights of the indicators are computed using the **Mon formula**, which integrates the AHP and CRITIC methods of indicator weighting. The resulting weights take into account both subjective and objective perspectives, enabling the calculation of a comprehensive weight vector of indicators based on **AHP-CRITIC**. The most important thing is the time for the formation of **cultural formation**. The weight is **0.3094**, that is the most important correlation and innovative standards. The weight results for all indicators are shown in **Table 4**.

For Task 3, to further validate our model, we selected four SDEs that were recently added or removed from the Olympic program and three SDEs that have been consistently present since the 1988 Olympics or earlier. The selected SDEs are: **Karate, Surfing, Baseball & Softball, Breaking, Table Tennis, Swimming, Archery**. We applied our model to rank these SDEs and used **TOPSIS** methods to further validate the results. This dual validation ensures that our model can reliably evaluate and recommend SDEs for future Olympic Games, aligning with the IOC's evolving standards and values.

For Task 4, we collect the indicator data of other largescale events SDES, use the TOPSIS method, calculate the SDES score and order ranking, and select the three SDES that are not available in the Olympic Games in order as the newly added or reintroduced SDES of the 2032 Olympic Games: **intellectual projects, boxing, martial arts (Add in order)**. Then the three new and the original SDEs of the Olympic Games will be recalculated and sorted. The top 30 SDEs can be used as the SDEs of the 2038 Olympic Games.

For Task 5 and Task 6, we make **sensitivity analysis** of the establishment of the **TOPSIS** and **entropy weight method**, set $\pm 10\%$ of the disturbance in the weight change analysis of the influencing factors, setting $\pm 10\%$ disturbance to SDES analysis in a single indicator, and setting $\pm 5\%$ disturbance in the data accuracy. Then, the conclusion model is **highly stable** and **applicable**, suitable for evaluation of Olympic sport. Finally,we write a letter addressed to the IOC summarizing our findings.

Keywords: Entropy Weight Method; AHP-CRITIC; TOPSIS; Olympic SDEs; Sensitivity analysis

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1 Introduction

1.1 Problem Background

The Olympics is always full of competition and excitement, and all of this is due to the numerous eye-catching sports, disciplines,or events(SDEs)in the Olympic Games. Every hosting nation have to schedule the SDEs before the Olympics, and we can evidently see the distribution of Olympic sports that have been held in Figure 1.

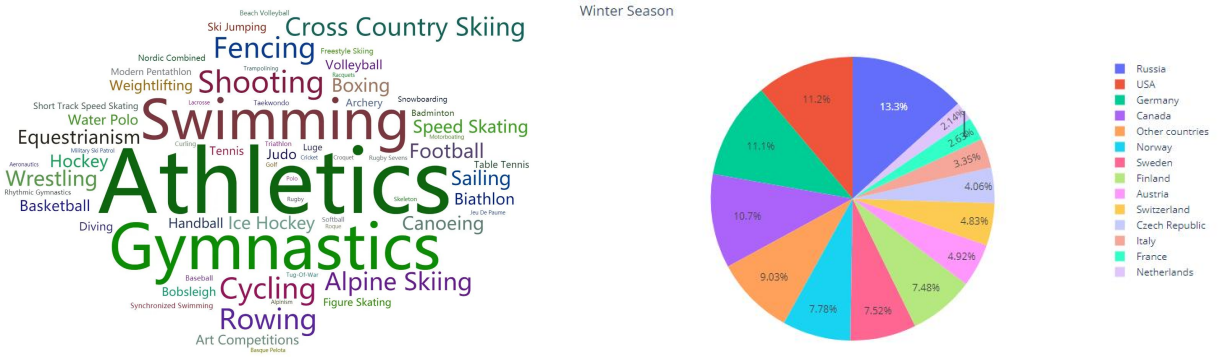

Figure 1. Distribution of Olympic sports that have been held

The data above comes from the website Minitab Blog. Every hosting nation wants to change the SDEs, because some new events can show the nation characteristics, reflect the times and appeal to global audiences, for example the 2024 Paris Olympics added breaking into the SDEs. So when every time a nation wants to add a SDEs, the new SDEs have to follow the IOC Criteria made by the IOC's Olympic Programme Commission. This criteria ensure the new SDEs have relationship with the Olympic value, and the criteria will help to promote the Olympic games have strong equality, and the criteria can allow the SDEs connect with the times and will increase the number of global audiences watching Olympic.

1.2 Restatement of the Problem

To assist the IOC in deciding which SDEs (Sports, Disciplines, Events) should be added or removed from the 2032 Summer Olympics, we need to select indicators that align with the IOC standards and establish a mathematical model. Therefore, our paper will contain these of the following:

- Identify the factors that should be considered when handling the IOC's project selection criteria.
- To assist the IOC in evaluating which SDEs (Sports, Disciplines, Events) meet Olympic standards, establish a mathematical model using the factors chosen.
- Select at least three SDEs that have been added or removed for the 2020, 2024, and 2028 Olympics, as well as at least three SDEs that have consistently appeared in Olympic programs. Test the model by selecting different sets of SDEs, and explain how the model evaluates these SDEs.

- Choose three SDEs that could be newly added or reintroduced to the 2032 Brisbane Olympics and rank them. Identify any special competitive sports that might participate in the 2036 or later Olympics.
- Conduct a sensitivity analysis on the model to ensure its stability, determine which aspects of the model lead to high scores for SDEs, and analyze whether these aspects reflect the strengths and weaknesses of the model.
- Write a 1-2 page letter to IOC explaining the basic principles of the model and the evaluation results in non-technical terms.

1.3 Our Work

In order to make the idea of this article more clear, the drawing work flowchart is shown in the Figure 2.

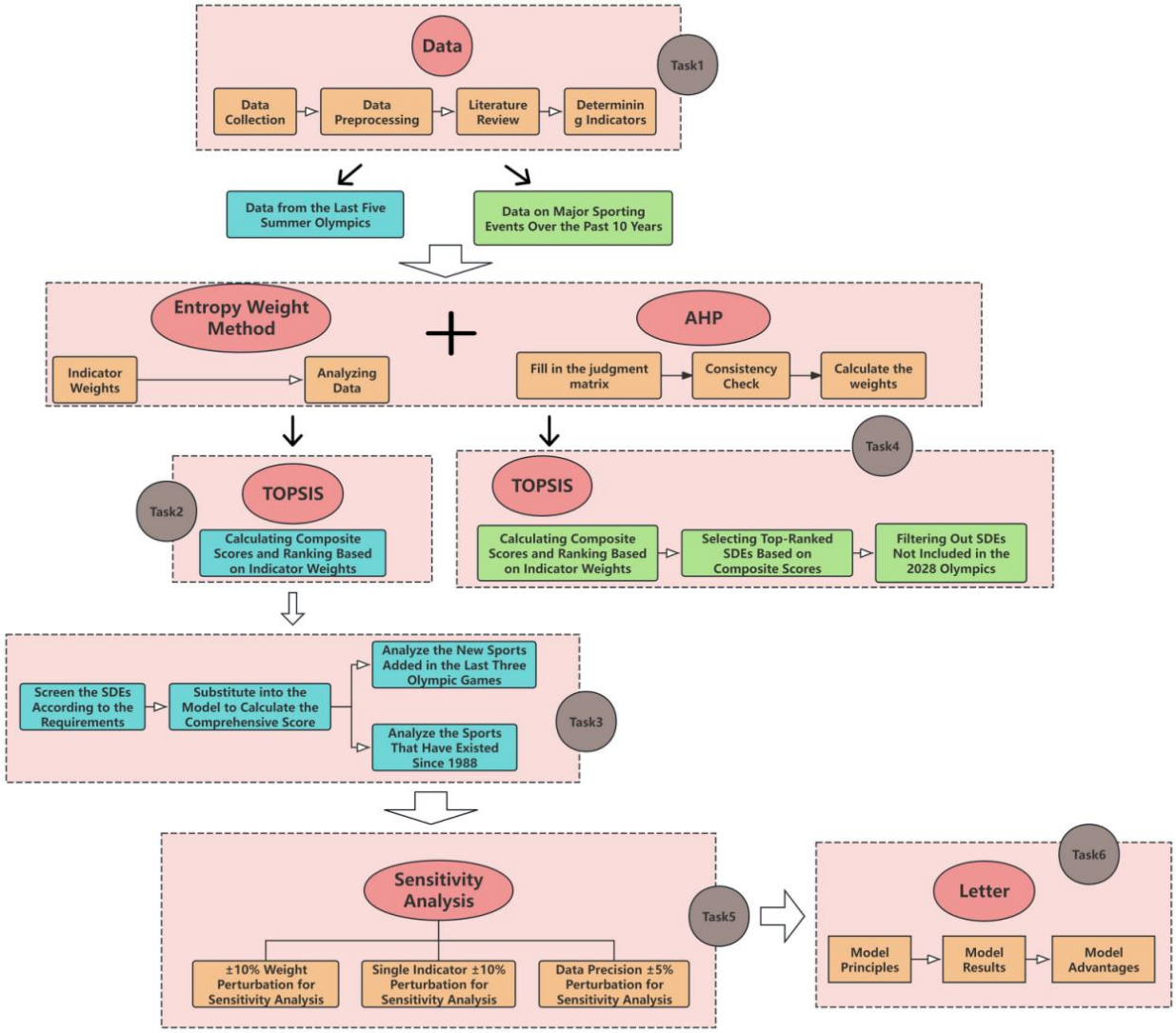

Figure 2. Workflow Diagram

2 Assumptions and Justifications

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

- **Assumption 1: Only major evaluation criteria influence event selection, ignoring other factors.**

  Justification: To simplify the model, we assume that only the primary evaluation criteria (popularity and accessibility, gender equity, sustainability, inclusivity, relevance and innovation, safety and fair play) influence the selection of events. other secondary factors are disregarded, and each criterion is considered independently.

- **Assumption 2: Current trends in sports popularity will continue.**

  Justification: We assume that the popularity trends observed for each sport over the past few years will generally continue through 2032.

- **Assumption 3: IOC Criteria will be applied consistently and without change in weighting.**

  Justification: We assume that the International Olympic Committee's(IOC) sport selection criteria will be applied consistently to all candidate sports without changing the emphasis placed on sporting events.

- **Assumption 4: Brisbane's infrastructure and cultural landscape influence sporting choices.**

  Justification: We assume that Brisbane's existing infrastructure and cultural landscape will play a role in the selection process. Sports that fit Brisbane's infrastructure capabilities, climate and local sporting interests are more likely to be considered for inclusion.

- **Assumption 5: Future social trends will influence how IOC standards are interpreted.**

  Justification: While the International Olympic Committee (IOC) standards are clearly defined, we believe that evolving social trends will influence how they are applied. For instance, sports that align with sustainability and digital innovation may be prioritized as these trends gain importance by 2032.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1. Notations used in this paper

Symbol	Description
z_{ij}	The variable after a normalized processing

$x_{\min}$	The minimum values of each indicator
$x_{\max}$	The maximum values of each indicator,
p_{ij}	The probability of item i under item j.
e_j	The information entropy of item j
d_j	The corresponding information effectiveness value
w_j	The entropy weight value of the j SDES evaluation indicator
D_i^+	The distance between the $i(i = 1,2,\cdots,n)$ evaluation object and the maximum value
D_i^-	The distance between the $i(i = 1,2,\cdots,n)$ evaluation object and the minimum value
s_i	the score of the i Olympic SED
S_i	the normalized score of the i Olympic SED

Note: There are some variables that are not listed here and will be discussed in detail in each section.

4 Model Preparation

4.1 Data Collection

To develop our model, we have gathered data from various reputable sources to ensure accuracy and reliability. These sources in Table 2 provide comprehensive and up-to-date information, enabling us to evaluate sports, disciplines, and events (SDEs) against the IOC's criteria effectively.

Table 2. Data source collation

Data Set	Website Source
Ratio of male to female participants (male:female)	https://unesdoc.unesco.org/
Number of doping violations	https://www.wada-ama.org/
Project injury rate (%)	https://pubmed.ncbi.nlm.nih.gov/
Amount of garbage generated (tons/session)	https://www.unep.org/
Input cost (million US dollars)	https://olympics.com/
Global popularity (1-5)	https://olympics.com/ioc/diversity-inclusion/
Average number of spectators (100 million)	https://www.nielsen.com/

4.2 Data Processing

We structured our data processing in several steps to ensure accurate, fair comparisons across sports, disciplines, and events (SDEs).Firstly, we collected data for each SDE relevant to our evaluation criteria, such as global participation, gender ratios, environmental impact, and country representation.Secondly, we applied Min-Max normalization to bring all criteria onto a consistent 0-1 scale, ensuring no single criterion overly influenced the results.Then, we assigned equal initial weights to each criterion. For sensitivity analysis, we

adjusted these weights to test how shifting IOC priorities (e.g., a focus on gender equality or sustainability) would affect rankings.Finally, we prepared the processed data for TOPSIS analysis, allowing us to calculate objective scores that reflect each SDE's alignment with IOC standards.

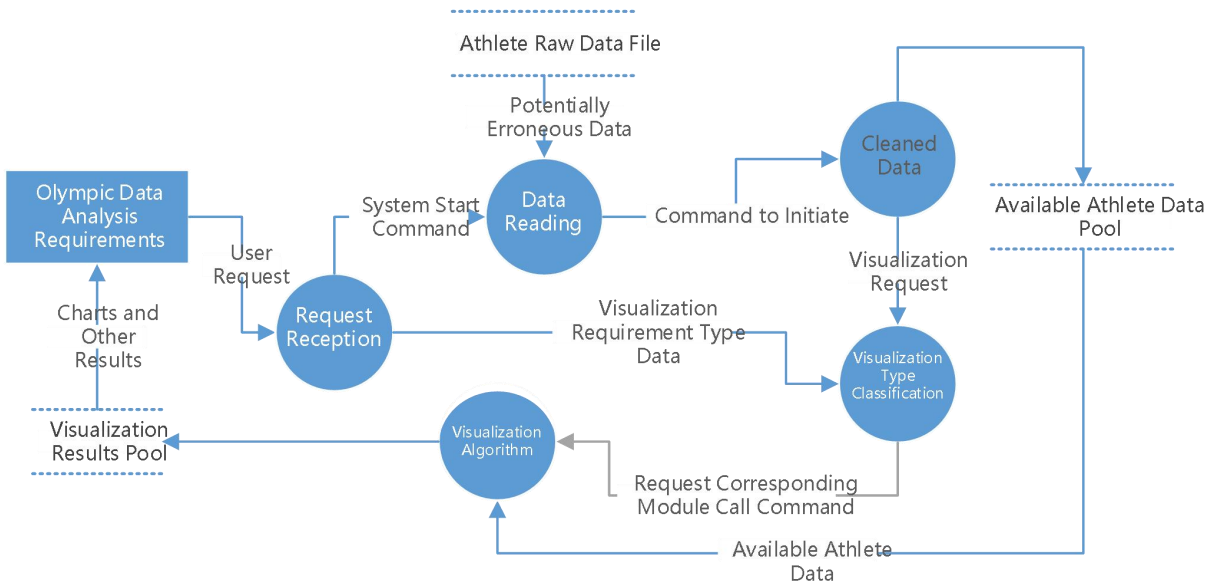

Figure 3. Data Processing Flowchart

4.3 Data Visualization

The Figure 4 reflects the trend of the number of men and women parameters of the Olympic Games over time. Among them, the red dot line indicates the number of male athletes, and the number of male athletes can fluctuate. The overall number of people is increasing but the number of people in some years will decline significantly, which may be related to the participating projects or social situations of the year. The black dot line indicates the number of female athletes. It can be found that the number of female athletes has risen steadily. The overall number of people increases with the increase of the year, and occasionally there will be a small fluctuation. The number of male athletes has always been more than female athletes. In recent sessions, the gap between the two is getting smaller and smaller, indicating that with the adjustment of the Olympic event, male and female athletes have obtained more equal opportunities.

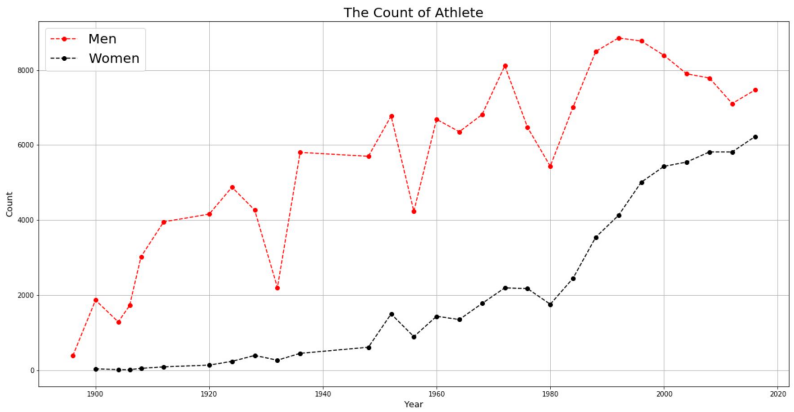

Figure 4. The Number of Male and Female Participants In the Olympics

The Athens Olympic Games in 1896 was the first summer Olympic Games. 311 athletes from the world participated, but women were excluded. Therefore, we can see that in the figure4 from the Olympic Games from 1900 to 1920, the number of female athletes is very small. In the Paris Olympics in 2024, the proportion of women's participation is about 50%. The efforts of female athletes and the lifting of the Olympic project broke through the invisible puppets.

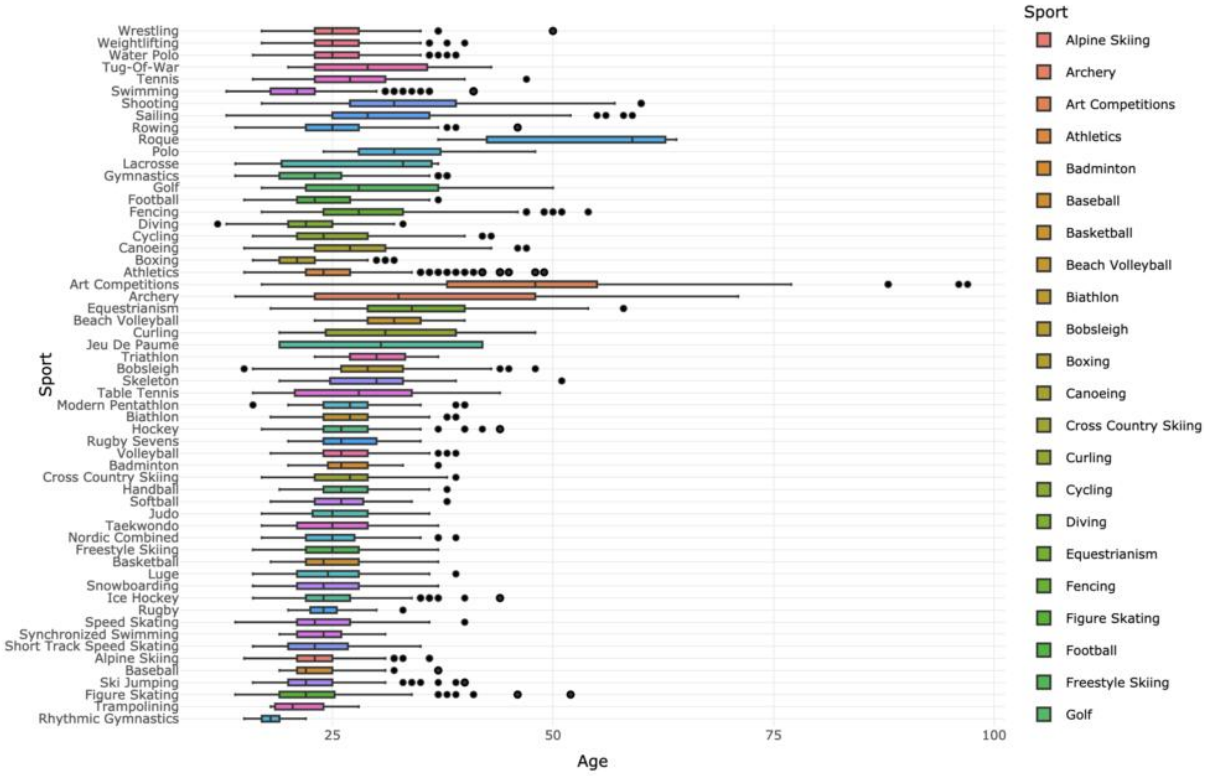

Figure 5. Age Residuals in Olympic Events Participation

It can be found from the Figure 5 that the age of the athletes is concentrated in 21-29 years. This is not only because young athletes occupy a certain advantage in physical strength, but

also reflect the attraction of young people from the side. More and more young athletes are willing to participate in the Olympic Games.

5 Evaluation Indicator System

Our system is designed to evaluate sports, disciplines, and events (SDEs) for the 2032 Summer Olympics based on the International Olympic Committee's (IOC) criteria. The system incorporates six main categories: gender equality, sustainability, inclusiveness, relevance and innovation, safety and fair play, and visibility and feasibility. Under each category, specific metrics are utilized to assess the alignment of SDEs with the IOC's objectives. For instance, gender equality is measured through the ratio of male to female participants; sustainability considers input costs and the amount of garbage generated; inclusiveness examines global popularity and the number of winning countries; relevance and innovation look at cultural formation time and the average age of athletes; safety and fair play focus on doping violations, injury rates, and health effects; and visibility and feasibility analyze understanding difficulty, spectator numbers, training periods, economic impacts, and tech coverage. By systematically analyzing these metrics, our system provides a comprehensive framework for recommending SDEs that best fit the evolving vision of the Olympics.

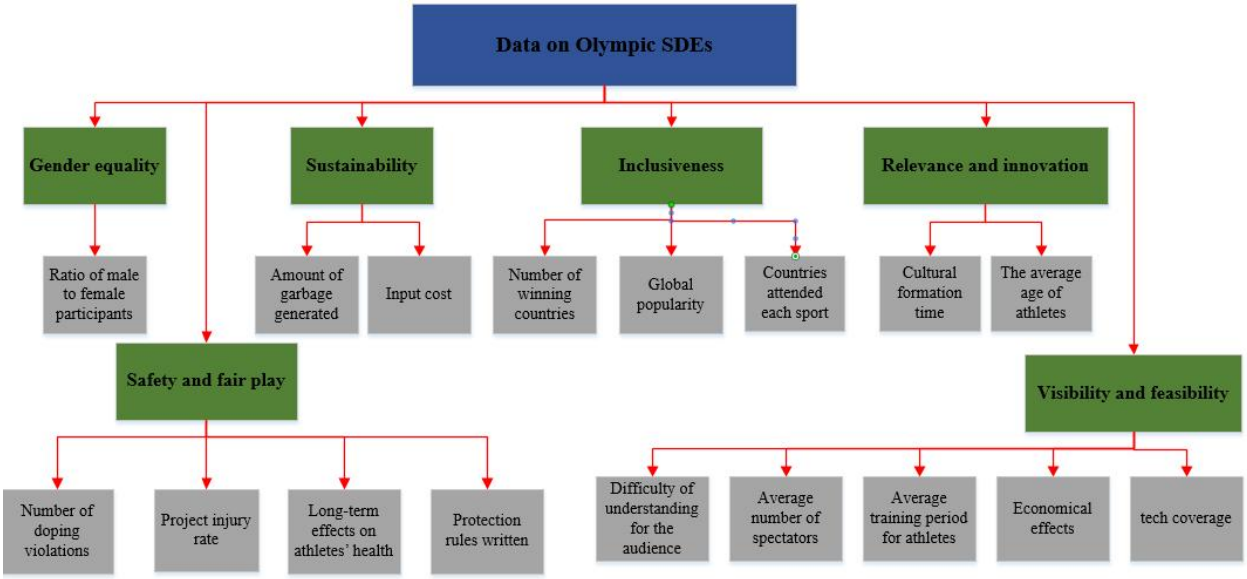

Figure 6. Hierarchical Chart of Indicators

5.1 *Gender equality(GE)*

- **Ratio of male to female participants.** The proportion of the number of men to women in the competition events. The proportion of the number of male athletes and female athletes can reflect the men's and women's equality of the competition project to a certain extent.

5.2 *Safety and fair play(SFP)*

- **Number of doping violations.** The number of people who violated the rules by using doping in competition events. The less the number of stimulants, the more explained the fairness of the competition project.
- **Project injury rate.** The ratio of the number of injured people in competition events to the total number of participants. Athlete's injury rate can reflect the safety of the competition items.
- **Long-term effects on athletes' health.** The degree of impact on the future health of athletes. Safe competition items have a positive and harmless impact on the long -term health of athletes.
- **Protection rules written.** The number of protection rules issued for competition events. The number of rules can reflect the fairness and security of the game to a certain extent.

5.3 *Sustainability (STB)*

- **Amount of garbage generated.** The weight of waste directly or indirectly generated after the project competitions. The less the garbage generated in the competition, the more it illustrates the sustainability of the competition items.
- **Input cost.** The cost invested in completing a competition. The lower the cost of the competition project, the more it shows that it can be held in sustainability.

5.4 *Inclusiveness (ICS)*

- **Number of winning countries.** The total number of countries where all the winners come from. The more the winning countries, the more the competition items are welcomed by the world.
- **Global popularity.** The degree of participation of countries around the world in competition events. The more times the competition items are held around the world, the more explain the popularity of the project.
- **Countries attended each sport.** The number of countries participating in each event. The more countries in registering for the Olympic Games, the more popular the project is.

5.5 *Relevance and innovation (RAI)*

- **Cultural formation time.** The time elapsed since the formation of the event. The earlier the formation of culture, the higher the degree of correlation with the world.
- **The average age of athletes.** The average age of the members of the sports team. The lower the average age athlete, it means that the project is more inclined to modern, and the higher the innovation.

5.6 *Visibility and feasibility (VAF)*

- **Difficulty of understanding for the audience.** The degree of recognition and understanding of competition events by the audience. The understanding of the audience's understanding of the competition project can reflect the feasibility of the project.

- **Average number of spectators.** The average number of people watching the competition both online and offline. The more people watch the competition items, the more they can explain the visibility and popularity of the competition items.
- **Average training period for athletes.** The training time required for each athlete to participate in the competition events. The shorter the athlete's training cycle, it means that the scientific and visible competition items.
- **Economical effects.** The impact of competition events on various aspects of the economy. The smaller the economic burden of the competition project, the higher the feasibility
- **Tech coverage.** The global scope covered by the project. The higher the area coverage of the competition items, the more it shows its feasibility at the sports meeting.

6 SDEs Evaluation Model Based on Combined Weight-TOPSIS

6.1 Data Pre -processing

Table 3. Olympic SEDs Evaluation Indicators

First level indicator	Secondary indicators	Indicator Types
Gender equality	Ratio of male to female participants (male:female)	Intermediate Type
Safety and fair play	Number of doping violations	-
	Project injury rate (%)	-
	Long-term effects on athletes' health (1-5)	-
	Protection rules written	+
Sustainability[1]	Amount of garbage generated (tons/session)	-
	Input cost (million US dollars)[2]	-
Inclusiveness	Number of winning countries	+
	Global popularity (1-5)	+
	Countries attended each sport	+
Relevance and innovation	Cultural formation time (years)	+
	The average age of athletes	+
Visibility and feasibility[3]	Difficulty of understanding for the audience (1-5)	-
	Average number of spectators (100 million)	+
	Average training period for athletes (years)	-
	Economical effects (money earned for government held)	+
	Tech coverage	+

"+" means Cost Type, "-" means Beneficial Type.

Cost Type: The greater the data, the more unfavorable it is for the comprehensive score, so it is necessary.

Beneficial Type: The greater the data value, the more beneficial it is for the results, so there is no need to deal with positiveization.

In order to eliminate the indicator units for data, we have standardized all data. Positive and standardized matrix to ensure the non-negativeness of each element in the matrix.

$$z_{ij} = \frac{x_{ij} - x_{min}}{x_{max} - x_{min}} \tag{1}$$

Among them, z_{ij} is the variable after a normalized processing, $x_{\min}$ and $x_{\max}$ are the maximum and minimum values of each indicator, respectively.

6.2 Determination of the Weights for Indicators

6.2.1 Analytic Hierarchy Process

First, a hierarchy diagram is constructed based on the previously selected metrics, as shown in Figure 7 below.

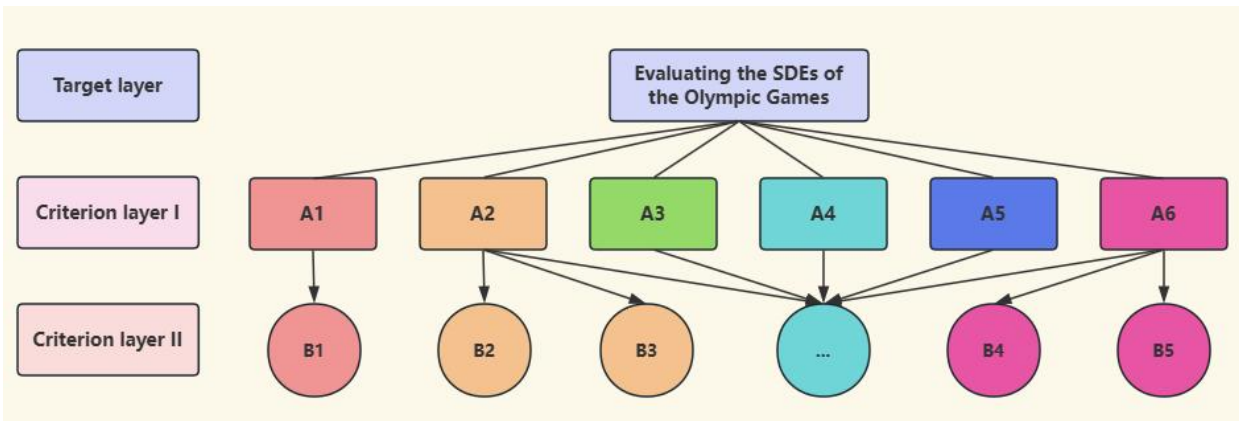

Figure 7. The hierarchical structure diagram of AHP

In the Figure 7, A1-A6 represent the six criteria for evaluating the SDEs (Sport Development Elements) of the Olympic Games provided by the IOC (International Olympic Committee), B1-B15 represent the 15 evaluation indicators selected in this paper, and SDE1-SDE33 represent the 33 events in the Olympics.

- **Step 1:Fill in the judgment matrix**

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \tag{2}$$

Among them, a_{ij} means elements in the matrix, A means judgment matrix.

- **Step 2:Consistency Check**

(1) Calculate the Consistency Index (CI)

$$CI = \frac{\lambda_{\max} - n}{n - 1} \tag{3}$$

Among them, CI means Consistency Index, $\lambda_{\max}$ means the maximum feature value of the matrix.

(2)Look up the corresponding Random Consistency Index (RI)

(3)Calculate the Consistency Ratio (CR)

$$CR = \frac{CI}{RI} \tag{4}$$

Among them, CR means Consistency Ratio. At this point, $CR < 0.1$, the consistency of the judgment matrix is acceptable.

- **Step 3:Calculate the weights**

(1) Calculate the weights using the arithmetic mean method

$$w_{1i} = \frac{1}{n}\sum_{j=1}^{n} \frac{a_{ij}}{\sum_{k=1}^{n} a_{kj}} \tag{5}$$

Among them, ω_{1i} means arithmetic mean weights.

(2) Calculate the weights using the geometric mean method

$$w_{2i} = \frac{\left(\prod_{j=1}^{n} a_{ij}\right)^{\frac{t}{n}}}{\sum_{k=1}^{n}\left(\prod_{j=1}^{n} a_{kj}\right)^{\frac{1}{n}}} \tag{6}$$

Among them, ω_{2i} means geometric mean weights.

(3)Calculate the weights using the eigenvalue method ω_{3i}

The final weights for evaluating the SDEs of the Olympic Games should be

$$\omega_i = \frac{1}{3}(\omega_{1i} + \omega_{2i} + \omega_{3i}). \tag{7}$$

6.2.2 Entropy Weight model

After reviewing the documents and collecting data, the 6 standards of IOC: gender equality, security and fair competition, sustainability, tolerance, correlation and innovation, popularity and feasibility are determined as first -level indicators. At the same time, 17 second -level indicators were found to reflect the first -level indicators. The specific indicators are shown in the Table 3.

Entropy weight model [4] is the concept of information entropy is a decision -making tool established. It is an objective method of empowerment. It can help us avoid interference of subjective factors and seek the weight of the evaluation SDES that meets the Olympic Standard Evaluation Index.

(1) Calculate the proportion of the i SED under item j evaluation index, and see it as the probability used in relative entropy calculations.

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^{n} z_{ij}} \tag{8}$$

Among them, p_{ij} represents the probability of item i under item j.
(2) Calculate the information entropy e_j of item j evaluation index, and calculate the corresponding information utility value d_j.

$$e_j = -\frac{1}{ln\, n}\sum_{i=1}^{n} p_{ij} ln\left(p_{ij}\right) \tag{9}$$

$$d_j = 1 - e_j \tag{10}$$

Among them, e_j represents the information entropy of item j, d_j represents the corresponding information effectiveness value.
(3) Normalize to obtain the entropy weight w_j of each evaluation indicator for the Olympic SDEs.

$$w_j = d_j / \sum_{j=1}^{m} d_j \tag{11}$$

Among them, w_j represents the entropy weight value of the j SDES evaluation indicator.

6.2.3 Weighted Combination of AHP and CRITIC

The final weights of the indicators are computed using the Mon formula, which integrates the AHP and CRITIC methods of indicator weighting. The resulting weights take into account both subjective and objective perspectives, enabling the calculation of a comprehensive weight vector of indicators based on AHP-CRITIC.

$$\omega_{\text{total}} = \beta \cdot \omega_{AHP} + (1 - \beta) \cdot \omega_{\text{CRITIC}}\ (0 \leqslant \beta \leqslant 1) \tag{12}$$

The weights of the 17 secondary indicators obtained are respectively.

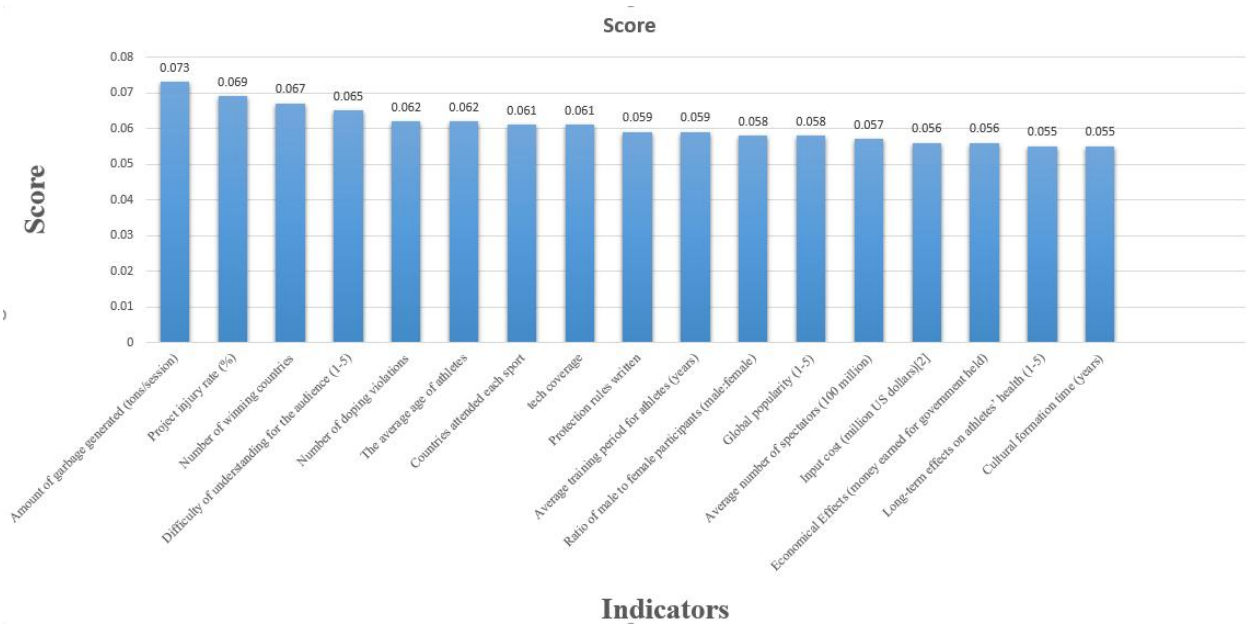

Figure 8. Index right to weight pillar chart

The specific weight of the income is shown in the Figure 8, and its specific value is as shown in the Table 4.

Table 4. Weights of Olympic SEDs Evaluation Indicators

Indicators	Score	**Rank**	Indicators	Score	**Rank**
Amount of garbage generated (tons/session)	0.073	**1**	Average training period for athletes (years)	0.059	**10**
Project injury rate (%)	0.069	**2**	Ratio of male to female participants (male:female)	0.058	**11**
Number of winning countries	0.067	**3**	Global popularity (1-5)	0.058	**12**
Difficulty of understanding for the audience (1-5)	0.065	**4**	Average number of spectators (100 million)	0.057	**13**
Number of doping violations	0.062	**5**	Input cost (million US dollars)[2]	0.056	**14**
The average age of athletes	0.062	**6**	Economical Effects (money earned for government held)	0.056	**15**
Countries attended each sport	0.061	**7**	Long-term effects on athletes' health (1-5)	0.055	**16**
tech coverage	0.061	**8**	Cultural formation time (years)	0.055	**17**
Protection rules written	0.059	**9**			

It can be seen from the table4 that in the indicator of the Olympic SDES, the most important thing is the time for the formation of cultural formation. The weight is 0.3094, that is, the most important correlation and innovative standards. The second is the average number of viewers, the difficulty of the audience's understanding, is the awareness and feasibility standards. The long -term impact on the health of the athletes is the most

important to evaluate the Olympic SDES evaluation. The investment cost and the weight of global popularization are less than 0.02. This article believes that the results of these two indicators have little impact. Therefore, discarding these two indicators, only the remaining 10 indicators are considered an important influencing factor for the Olympic SDEs assessment.

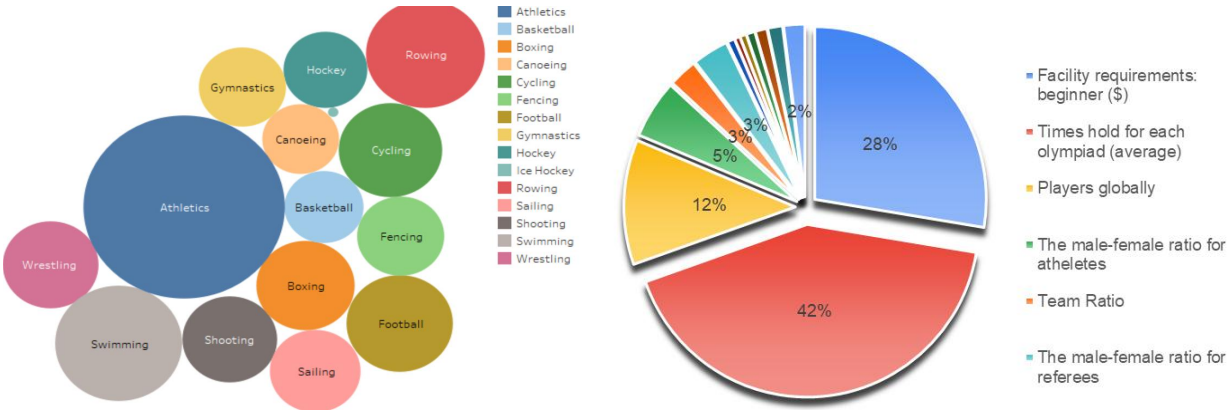

Figure 9. Weight Proportion Chart of Olympic Events

6.3 TOPSIS Model

The TOPSIS algorithm was proposed to address multi-objective decision-making problems with limited alternatives in systems engineering. It is a distance-based comprehensive evaluation method. In recent years, this algorithm has been applied to various fields such as risk decision analysis, evaluation of land consolidation implementation benefits, and performance assessment [5].

(1) To define the maximum and minimum values

Find out the maximum value of each Olympic SDES evaluation indicator, remember as $z_i^+(i = 1,2,\cdots,m)$, form a vector $Z^+ = \{z_1^+, z_2^+, \cdots, z_m^+\}$, this vector represents the most ideal Olympic SED value.

Find out the minimum value of each Olympic SDES evaluation indicator, remember as $z_i^-(i = 1,2,\cdots,m)$, form a vector $Z^- = \{z_1^-, z_2^-, \cdots, z_m^-\}$, this vector represents the most unsatisfactory Olympic SED value.

(2) The distance between definition and maximum and minimum values

Define the distance between the $i(i = 1,2,\cdots,n)$ evaluation object and the maximum value is

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \left(z_j^+ - z_{ij}\right)^2} \tag{13}$$

Define the distance between the $i(i = 1,2,\cdots,n)$ evaluation object and the minimum value is

$$D_i^- = \sqrt{\sum_{j=1}^{m} \left(z_j^- - z_{ij}\right)^2} \tag{14}$$

(3)Calculate the score of the i Olympic SED to s_i, and the calculation formula is

$$s_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{15}$$

(4)Calculate the normalized score S_i of the i Olympic SED. The calculation formula is

$$S_i = \frac{s_i}{\sum_{i=1}^{n} s_i} \times 10000 \tag{16}$$

6.4 Olympic SDES evaluation scores and rankings

Skateboarding was added as a new event starting in 2020, ranking 19th in terms of score, placing it in the middle. This indicates that the inclusion of skateboarding in the Olympic Games is entirely reasonable because it meets the IOC's evaluation criteria. Field hockey is set to be introduced as a new event in the 2028 Olympics, ranking 28th in score, which is at the bottom. This event will replace some existing Olympic sports, but due to its first appearance in the Olympics, its score is still relatively low. Whether it will continue to be part of the Olympic program after 2028 will depend on audience reception following the Games.

Karate ranks 21st, indicating a lower position. This sport only appeared in the 2020 Tokyo Olympics as a special event for the host country Japan, suggesting that while it has strong cultural characteristics, it lacks inclusivity and is not a globally popular or widely promoted sport.

Table 5. Olympic SEDs scores and rankings

Sports	Scores	Ranking	Sports	Scores	Ranking
Football	0.0634	1	Judo	0.0271	18
Athletics	0.0594	2	Skateboarding	0.0271	19
Aquatics	0.0497	3	Handball	0.0268	20
Archery	0.0434	4	Karate	0.0261	21
Gymnastics	0.0425	5	Golf	0.0258	22
Boxing	0.0424	6	Cycling	0.024	23
Equestrian	0.0419	7	Taekwondo	0.0226	24
Wrestling	0.0361	8	Baseball and Softball	0.0224	25
Basketball	0.0328	9	Surfing	0.0217	26
Table Tennis	0.0328	10	Field Hockey	0.0216	27
Tennis	0.0319	11	Sport Climbing	0.0213	28
Volleyball	0.0291	12	Rugby	0.0206	29
Weightlifting	0.0289	13	Canoeing	0.0202	30

Fencing	0.0275	14	Rowing	0.0179	31
Sailing	0.0275	15	Modern Pentathlon	0.017	32
Badminton	0.0272	16	Triathlon	0.0142	33
Shooting	0.0272	17			

7 Model Application and Test

7.1 Calculate other large events SDES scores

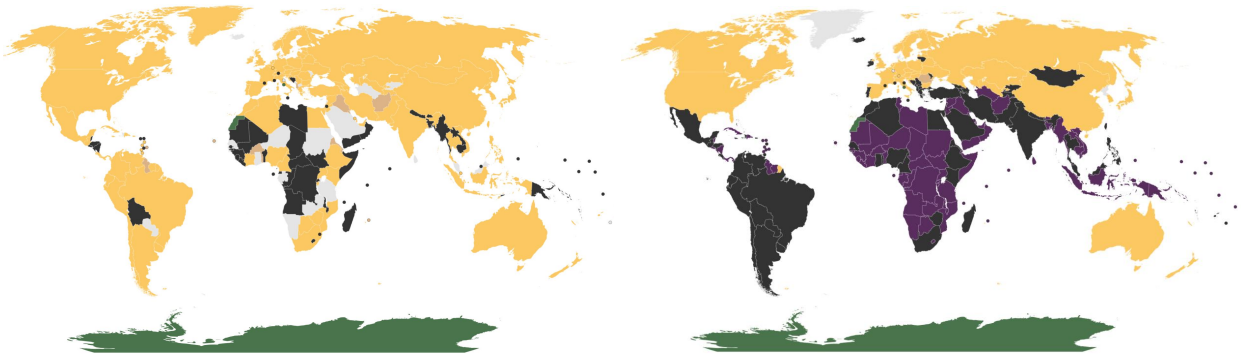

Figure 10. Map Showing the Distribution of Countries with Medal Winners

This paper has gathered SDEs data from the Paralympics, African Games, Asian Games, and European Games, and input this data into Model Two. By applying the TOPSIS method again, we can obtain the scores for SDEs of other major sporting events.

7.2 Added or Removed SDEs

Table 6. SEDs scores and rankings in other events

Sports	Scores	Ranking	Sports	Scores	Ranking
Aquatics	0.0761	1	Tennis	0.0202	22
Intellectual Project	0.0652	2	Volleyball	0.0202	23
Aquatics	0.0589	3	Kayaking	0.0199	24
Swordplay	0.0423	4	Fencing	0.0196	25
Boxing	0.0319	5	Dragon Boat	0.0192	26
Martial Arts	0.0274	6	Cricket	0.0191	27
Football	0.0254	7	Sailboat	0.0185	28
Shooting	0.0251	8	Biking	0.0184	29
Rowing	0.0241	9	Weightlifting	0.0184	30
Ping-pong	0.0233	10	Wrestling	0.0183	31
Sepak Takraw	0.0231	11	Basketball	0.0181	32
Badminton	0.023	12	Gymnastics	0.018	33
Equestrian	0.0225	13	Taekwondo	0.0177	34
E-Sports	0.0221	14	Modern Pentathlon	0.0176	35
Breakdance	0.022	15	Baseball and Softball	0.0174	36
Squash	0.022	16	Handball	0.0172	37
Roller skating	0.0214	17	Kabaddi	0.0171	38

Golf	0.0209	18	Hockey	0.0162	39
Judo	0.0207	19	Triathlon	0.016	40
Climbing	0.0205	20	Rugby	0.0149	41
Archery	0.0202	21			

Among the SDEs of other major sporting events, the top three projects that are not currently part of the Olympics but have the highest scores in these events are ranked as follows: Intellectual Sports at 2nd place, Wushu at 4th place, and Boxing at 5th place. Therefore, this paper suggests that Intellectual Sports, Wushu, and Boxing should be considered as new additions to the 2032 Olympic Games. However, it's worth noting that Boxing is already a part of the Olympic Games, so the suggestion might specifically refer to enhancing or adjusting the current format or inclusion of Boxing, or it could be a typographical error in the original text. If the intention is to suggest new sports, the recommendation would apply to Intellectual Sports and Wushu, with clarification needed regarding Boxing.

7.3 Continuous SDEs and Potential SDEs

Here is the ranking of 36 Sport Discipline Events after integrating the three new sports for the 2032 Olympics with the existing Olympic sports:

Table 7. 2038 Olympic Games SEDs Scores Ranking

Sports	Scores	Ranking	Sports	Scores	Ranking
Football	0.0607	1	Shooting	0.0247	19
Athletics	0.0551	2	Karate	0.0234	20
Aquatics	0.0452	3	Golf	0.023	21
Gymnastics	0.0442	4	Cycling	0.0229	22
Equestrian	0.0405	5	Volleyball	0.0228	23
Archery	0.0402	6	Baseball and Softball	0.0204	24
Boxing	0.0395	7	Taekwondo	0.0201	25
Intellectual Project	0.0389	8	Skateboarding	0.0201	26
Wrestling	0.0335	9	Canoeing	0.02	27
Tennis	0.0302	10	Boxing	0.0196	28
Swordplay	0.0285	11	Surfing	0.0193	29
Basketball	0.0282	12	Field Hockey	0.0192	30
Table Tennis	0.0271	13	Sport Climbing	0.0189	31
Sailing	0.0266	14	Handball	0.0183	32
Weightlifting	0.0265	15	Rugby	0.0183	33
Badminton	0.0252	16	Rowing	0.0181	34
Judo	0.0249	17	Modern Pentathlon	0.0167	35
Fencing	0.0248	18	Triathlon	0.0143	36

The top three sports with the highest scores, ranking 1st to 3rd, are **football**, **athletics**, and **swimming**. These three sports have been present since 1988 and continue to be included, demonstrating that they meet the IOC's standards for Olympic events. This also indirectly suggests that the model used for evaluation is robust and reliable.

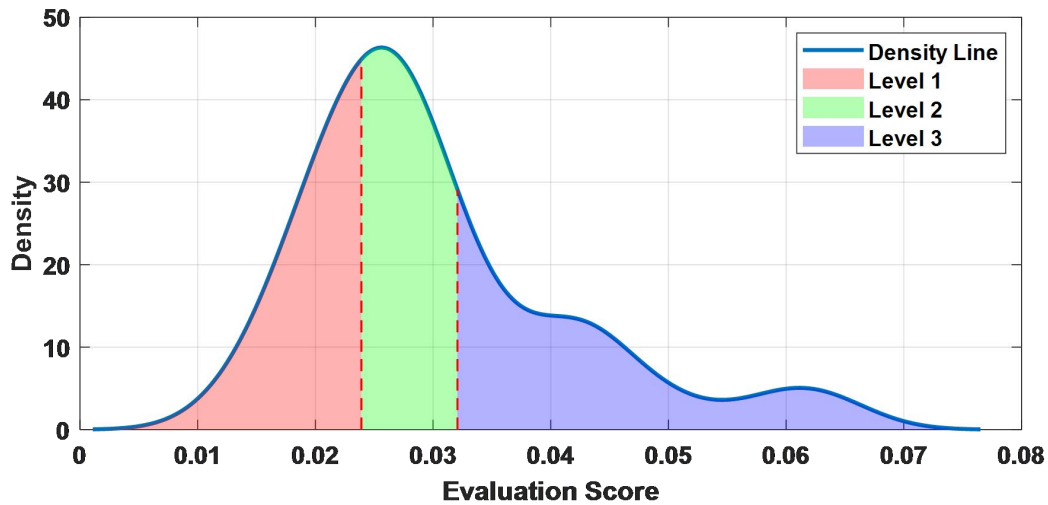

Figure 11. SDEs level classification diagram

Because the Olympic program typically includes more than thirty events, this article considers that **the top 30 Sport Discipline Events** in Table 7 have a high probability of becoming SDEs for the 2038 Olympics.

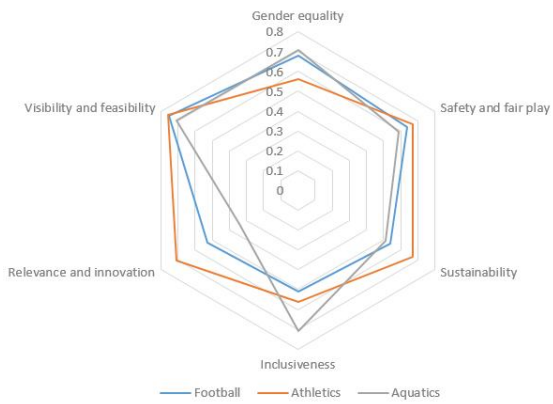

Figure 12. Radar Chart for Scoring Football, Athletics and Swimming

8 Sensitivity Analysis

In the previous section, this paper established a TOPSIS entropy weight method model to comprehensively evaluate various sports. To understand the robustness of the model and its sensitivity to changes in different input parameters, and to assess the credibility and adaptability of the model, this paper will use sensitivity analysis for validation.

In the sensitivity analysis, we will discuss the model's robustness and sensitivity to changes in input parameters from three perspectives: weight sensitivity analysis, single indicator sensitivity analysis, and data precision sensitivity analysis.

8.1 Weight Sensitivity Analysis

First, for the weight changes, we will introduce ±10% perturbations to each weight individually and recalculate the scores for each SDE. The results are shown in Figure 14.

Blue Line: Initial scores using the baseline weights.

Orange Line: Scores after increasing each criterion's weight by 10%.

Purple Line: Scores after decreasing each criterion's weight by 10%.

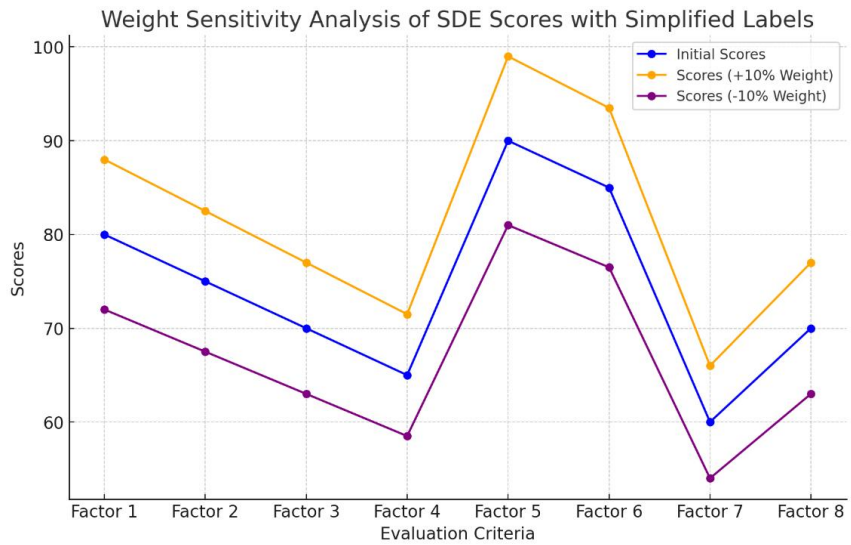

Figure 13. Sensitivity Analysis of Weight Changes

From the results shown in Figure 13, it can be observed that the changes in weights have a relatively stable impact on the SDE scores, indicating that our model is robust to weight variations.

8.2 Single Criteria Sensitivity Analysis

For a single criterion sensitivity analysis, the values of each criterion were adjusted one by one, increasing and decreasing by 5%, respectively, and the score for each SDE was recalculated.

Using the scores above, we used the metric with the highest weighting score, "time of cultural formation", as a criterion, to increase and decrease by 5%, respectively, and then recalculated the score and ranking. Observe changes in scores and rankings. The results are shown in Figure15, where the yellow line represents the initial standard score, the green line represents the score when the criterion increases by 5%, and the red line indicates the score when the criterion decreases by 5%.

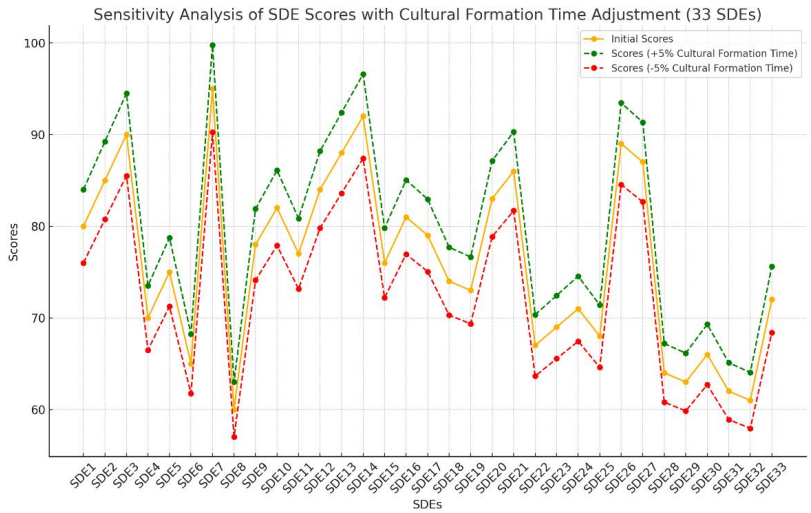

Figure 14. Single Criteria Sensitivity Analysis

As can be seen from Figure 14, the scores of Athletics, Boxing, and Gymnastics change significantly when the weight of "time to cultural formation" increases or decreases by 5%. This indicates that they are highly sensitive to this criterion and that these SDEs are more dependent on their historical and cultural value.

Badminton, Field Hockey, and Sport Climbing showed less dependence on this criterion by having less fluctuations in scores when the "time of cultural formation" weight changed. This means that such SDEs show strong robustness, indicating that they perform evenly across other standards and do not rely on a single historical and cultural context.

8.3 Data accuracy sensitivity analysis

For the sensitivity of data accuracy, this paper performs a ±5% perturbation of each input index, and recalculates the score of each SDE after each data perturbation. The results are shown in Figure 15.

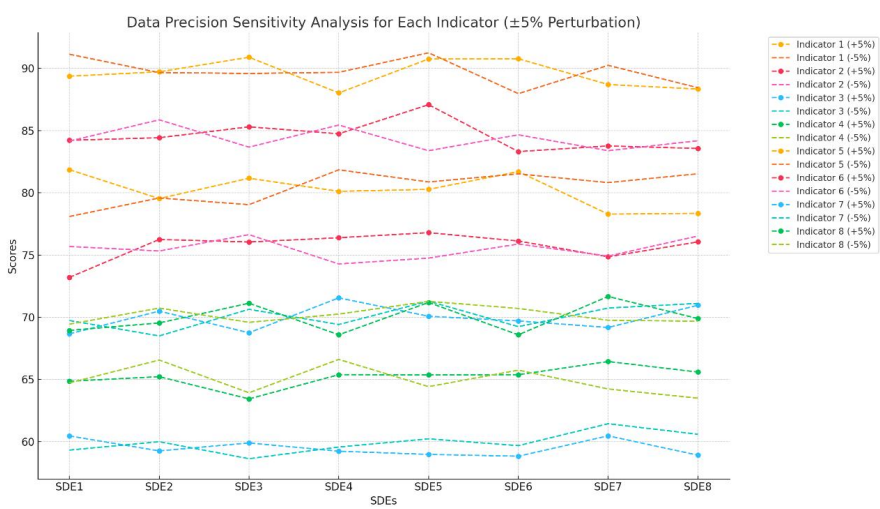

Figure 15. Data accuracy sensitivity analysis

Perturbations in average spectatorship and input costs can be observed to cause significant scoring variations across multiple SDEs, especially in events such as Athletics and Baseball and Softball. This indicates that the model has a high dependency on these metrics.

Number of winning countries and number of doping violations. Whether the perturbation is increased or decreased by 5%, the score for multiple SDEs varies less. This indicates that the model is less dependent on these metrics.

9 Evaluation of Strengths and Weaknesses

9.1 Strengths

- A large amount of data was used to verify the model and ensure the robustness of the model.
- This paper takes into account the Gender equality, Safety and fair play, Sustainability, Inclusiveness, Relevance and innovation, Visibility and feasibility aspects of the Olympic Games SDEs, and uses nine 17 indicators for quantitative calculations. The evaluation indicator system is well constructed.
- The entropy weight method is based on the concept of information entropy, which is an established decision-making tool, and is an objective empowerment method, which helps us to avoid the interference of subjective factors.

9.2 Weaknesses and Further Improvements

- Other models, such as principal component analysis, can be used for analysis, and then the results of multiple models can be compared to select the optimal model.
- The international political environment is not fully considered in this paper. It is difficult to predict and quantify because it involves international relations, social networks, etc.

10 Conclusion

The model shows good robustness to the weight changes of multiple criteria, and a high tolerance for small data errors in most indicators. This means that even if there is a slight error in the input data, it is still possible to maintain a high degree of evaluation stability and result accuracy. This provides a reliability guarantee for the application and promotion of the model. The disadvantage is the dependence on certain key indicators, such as the number of spectators and input costs. Therefore, in modern replication decision-making scenarios, the standard weight configuration may need to be further optimized. In summary, the TOPSIS entropy weight model has high robustness and applicability in evaluating SDEs.

11 Letter addressed to the IOC

To: International Olympic Committee

From: HIMCM TEAM #15292

Date:November 19, 2024

Subject: To Play or not to Play: Modeling Future Olympic Games

Dear Olympics Programme Commission of the IOC,

It is a great honor to have been hired to assist in the assessment of the SDEs for the Brisbane 2032 Olympic Games. Here, we would like to report to you in a more understandable way the results of our research, including the basic principles of the model we have built, as well as the results and recommendations of the evaluation of various SDEs.

We fully understand the importance of the Olympic Games to global sports and that every SDE selected for the Olympics must comprehensively align with the core values of Olympism. Based on this, when constructing our evaluation model, we considered numerous key factors designed to holistically measure whether each SDE is suitable for inclusion in the Olympics. We integrated these factors and assigned appropriate weights to each using certain mathematical methods, thereby building a model capable of quantitatively assessing SDEs.

Based on the model we constructed, we evaluated a series of SDEs. Organizing skateboarding competitions is relatively flexible in terms of venue setup, and the costs are more manageable. In terms of gender equality, both male and female athletes can participate equally at all levels of competition. Regarding sustainability, as environmental awareness spreads, the manufacturing of skateboarding equipment increasingly emphasizes the use of eco-friendly materials. Its innovative competition format has attracted a large number of young viewers, effectively reflecting relevance and innovation. Moreover, in terms of safety and fair play, there are well-established rules to ensure these aspects. Therefore, through our model evaluation, the skateboarding event scores high across all factors, which explains why it was successfully included in the 2020 Tokyo Olympics and has been widely welcomed.

Based on the model evaluation, we also recommend the following three SDEs that have the potential to be added or reintroduced to the 2032 Brisbane Olympics, listed in order of priority: Intellectual Sports, Wushu, and Boxing. The model we have constructed evaluates different SDEs by comprehensively considering multiple key factors. Through the analysis of recent Olympic-related events and the recommendations for potential events for the 2032 Brisbane Olympics, it can be seen that the model is capable of reflecting, to a certain extent, how well each SDE aligns with Olympic standards. We hope that the above analysis and recommendations can provide some valuable references for the International Olympic Committee (IOC) when deciding on the event selections for the 2032 Brisbane Olympics and future Olympic Games.

Thank you once again for giving us this valuable opportunity to collaborate!

Yours Sincerely,

HIMCM TEAM #15292

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